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The Practice of Medicine

The Editors

ENDURING VALUES OF THE MEDICAL PROFESSION

No greater opportunity, responsibility, or obligation can fall to the lot of a human being than to become a physician. In the care of the suffering, [the physician] needs technical skill, scientific knowledge, and human understanding. Tact, sympathy, and understanding are expected of the physician, for the patient is no mere collection of symptoms, signs, disordered functions, damaged organs, and disturbed emotions. [The patient] is human, fearful, and hopeful, seeking relief, help, and reassurance.

—Harrison's Principles of Internal Medicine, 1950

The practice of medicine has changed in significant ways since the first edition of this book was published in 1950. The advent of molecular genetics, sophisticated new imaging techniques, robotics, and advances in bioinformatics and information technology have contributed to an explosion of scientific information that has changed fundamentally the way physicians define, diagnose, treat, and attempt to prevent disease. This growth of scientific knowledge continues to evolve at an accelerated pace.

The widespread use of electronic medical records and the Internet have altered the way physicians and other health care providers access and exchange information as a routine part of medical education and practice (Fig. 1-1). As today's physicians strive to integrate an ever-expanding body of scientific knowledge into everyday practice, it is critically important to remember two key principles: first, the ultimate goal of medicine is to prevent disease and, when it occurs, to diagnose it early and provide effective treatment; and second, despite 70 years of scientific advances since the first edition of this text, a trusting relationship between physician and patient still lies at the heart of effective patient care.

■ THE SCIENCE AND ART OF MEDICINE

Deductive reasoning and applied technology form the foundation for the approach and solution to many clinical problems. Extraordinary advances in biochemistry, cell biology, immunology, and genomics,



FIGURE 1-1 *The Doctor* by Luke Fildes depicts the caring relationship between this Victorian physician and a very ill child. Painted in 1891, the painting reflects the death of the painter's young son from typhoid fever and was intended to reflect the compassionate care provided by the physician even when his tools were not able to influence the course of disease. (Source: History and Art Collection/Alamy Stock Photo.)

coupled with newly developed imaging techniques, provide a window into the most remote recesses of the body and allow access to the innermost parts of the cell. Revelations about the nature of genes and single cells have opened a portal for understanding physiology and unraveling the causes of disease. Researchers are deciphering the complex mechanisms by which genes are regulated, and increasingly, physicians are learning how subtle changes in many different genes, acting in an integrative contextual way, can affect the function of cells and organisms. Clinicians have developed a new appreciation of the role of stem cells in the renewal and repair of normal tissues, in the development of cancer and other disorders, and in the treatment of certain diseases. Entirely new areas of research, including studies of the human microbiome, epigenetics, and noncoding RNAs as regulatory features of the genome, have become important for understanding both health and disease. Extraordinary advances in vaccine platform technology and the use of cryo-electron microscopy for the structure-based design of vaccine immunogens have transformed the field of vaccinology, resulting in the unprecedented speed and success with which COVID-19 vaccines were developed. Information technology enables the interrogation of medical records from millions of individuals, yielding new insights into the etiology, characteristics, prognosis, and stratification of many diseases. With the increasing availability of very large data sets ("big data") from omic analyses and the electronic medical record, there is now a growing interest in machine learning and artificial intelligence for unbiased analyses that enhance clinical predictive accuracy. The field of artificial intelligence in particular is expanding rapidly, perhaps more rapidly than our understanding of how to use it optimally. In the coming years, it will be essential to learn the value and limitations of artificial intelligence methods and not abdicate decision-making entirely to a computer algorithm. The knowledge gleaned from the *science of medicine* continues to enhance the understanding by physicians of complex pathologic processes and to provide new approaches to disease prevention, diagnosis, and treatment. With continued refinement of unique omic signatures coupled with nuanced clinical pathophenotypes, the profession moves ever closer to practical precision medicine. Yet, skill in the most sophisticated applications of laboratory technology and in the use of the latest therapeutic modality alone does not make a good physician.

When a patient poses challenging clinical problems, an effective physician must be able to identify the crucial elements in a complex history and physical examination; order the appropriate laboratory, imaging, and diagnostic tests; and extract the key results from densely populated computer screens to determine whether to treat or to "watch." As the number of tests increases, so does the likelihood that some incidental finding, completely unrelated to the clinical problem at hand, will be uncovered. Deciding whether a clinical clue is worth pursuing or should be dismissed as a "red herring" and weighing whether a proposed test, preventive measure, or treatment entails a greater risk than the disease itself are essential judgments that a skilled clinician must make many times each day. This combination of medical knowledge, intuition, experience, and judgment defines the *art of medicine*, which is as necessary to the practice of medicine and the precision medicine of the future as is a sound scientific base, and as important for contemporary medical practice as it has been in earlier eras.

■ CLINICAL SKILLS

History-Taking The recorded history of an illness should include all the facts of medical significance in the life of the patient. Recent events should be given the most attention. Patients should, at some early point, be given the opportunity to tell their own story of the illness without frequent interruption and, when appropriate, should receive expressions of interest, encouragement, and empathy from the physician. Any event related by a patient, however trivial or seemingly irrelevant, may provide the key to solving the medical problem. A methodical review of systems is important to elicit features of an

underlying disease that might not be mentioned in the patient's narrative. In general, patients who feel comfortable with the physician will offer more complete information; thus, putting the patient at ease contributes substantially to obtaining an adequate history.

An informative history involves more than eliciting an orderly listing of symptoms. By listening to patients and noting the ways in which they describe their symptoms, physicians can gain valuable insight. Inflections of voice, facial expression, gestures, and attitude (i.e., "body language") may offer important clues to patients' perception of and reaction to their symptoms. Because patients vary considerably in their medical sophistication and ability to recall facts, the reported medical history should be corroborated whenever possible. The social history also can provide important insights into the types of diseases that should be considered and can identify practical considerations for subsequent management. The family history not only identifies rare genetic disorders or common exposures, but often reveals risk factors for common disorders, such as coronary heart disease, hypertension, autoimmunity, certain cancers, and asthma. A thorough family history may require input from multiple relatives to ensure completeness and accuracy. An experienced clinician can usually formulate a relevant differential diagnosis from the history alone, using the physical examination and diagnostic tests to narrow the list or reveal unexpected findings that lead to more focused inquiry.

The very act of eliciting the history provides the physician with an opportunity to establish or enhance a unique bond that can form the basis for a good patient–physician relationship. This process helps the physician develop an appreciation of the patient's view of the illness, the patient's expectations of the physician and the health care system, and the financial and social implications of the illness for the patient. Although current health care settings may impose time constraints on patient visits, it is important not to rush the encounter. A hurried approach may lead patients to believe that what they are relating is not of importance to the physician, and, as a result, they may withhold relevant information. The confidentiality of the patient–physician relationship cannot be overemphasized.

Physical Examination The purpose of the physical examination is to identify physical signs of disease. The significance of these objective indications of disease is enhanced when they confirm a functional or structural change already suggested by the patient's history. At times, however, physical signs may be the only evidence of disease and may not have been suggested by the history.

The physical examination should be methodical and thorough, with consideration given to the patient's comfort and modesty. Although attention is often directed by the history to the diseased organ or part of the body, the examination of a new patient must extend from head to toe in an objective search for abnormalities or other diagnostic clues. The results of the examination, like the details of the history, should be recorded at the time they are elicited—not hours later, when they are subject to the distortions of memory. Physical examination skills should be learned under direct observation of experienced clinicians. Even highly experienced clinicians can benefit from ongoing coaching and feedback. Simulation laboratories and standardized patients play an increasingly important role in the development of clinical skills. Although the skills of physical diagnosis are acquired with experience, it is not merely technique that determines success in identifying signs of disease. The detection of a few scattered petechiae, a faint diastolic murmur, or a small mass in the abdomen is not only a question of keen eyes and ears or more sensitive fingers but also of a mind alert to those findings. Because physical findings can change with time, the physical examination should be repeated as frequently as the clinical situation warrants.

Given the many highly sensitive diagnostic tests now available (particularly imaging techniques), it may be tempting to place less emphasis on the physical examination. Some are critical of physical diagnosis based on perceived low levels of specificity and sensitivity. Indeed, many patients are seen by consultants only after a series of diagnostic tests have been performed and the results are known. This fact should not deter the physician from performing a thorough

physical examination since important clinical findings may have escaped detection by diagnostic tests. Especially important, a thorough and thoughtful physical examination may render a laboratory finding unimportant (i.e., certain echocardiographic regurgitant lesions). The act of a hands-on examination of the patient also offers an opportunity for communication and may have reassuring effects that foster the patient–physician relationship.

Diagnostic Studies Physicians rely increasingly on a wide array of laboratory and imaging tests to make diagnoses and ultimately to solve clinical problems; however, such information does not relieve the physician from the responsibility of carefully observing and examining the patient. It is also essential to appreciate the limitations of diagnostic tests. By virtue of their apparent precision, these tests often gain an aura of certainty regardless of the fallibility of the tests themselves, the instruments used in the tests, and the individuals performing or interpreting the tests. Physicians must weigh the expense involved in laboratory procedures against the value of the information these procedures are likely to provide.

Single laboratory tests are rarely ordered. Instead, physicians generally request "batteries" of multiple tests, which often prove useful and can be performed with a single specimen at relatively low cost. For example, abnormalities of hepatic function may provide the clue to nonspecific symptoms such as generalized weakness and increased fatigability, suggesting a diagnosis of chronic liver disease. Sometimes a single abnormality, such as an elevated serum calcium level, points to a particular disease, such as hyperparathyroidism.

The thoughtful use of screening tests (e.g., measurement of low-density lipoprotein cholesterol) may allow early intervention to prevent disease (Chap. 6). Screening tests are most informative when they are directed toward common diseases and when their results indicate whether other potentially useful—but often costly—tests or interventions are needed. On the one hand, biochemical measurements, together with simple laboratory determinations such as routine serum chemistries, blood counts, and urinalysis, often provide a major clue to the presence of a pathologic process. On the other hand, the physician must learn to evaluate occasional screening-test abnormalities that do not necessarily connote significant disease. An in-depth workup after the report of an isolated laboratory abnormality in a person who is otherwise well is often wasteful and unproductive. Because so many tests are performed routinely for screening purposes, it is not unusual for one or two values to be slightly outside the normal range or to reflect physiologic variation. Nevertheless, even if there is no reason to suspect an underlying illness, tests yielding abnormal results ordinarily are repeated to rule out laboratory error. If an abnormality is confirmed, it is important to consider its potential significance in the context of the patient's condition and other test results.

There is almost continual development of technically improved imaging studies with greater sensitivity and specificity. These tests provide remarkably detailed anatomic information that can be pivotal in informing medical decision-making. MRI, CT, ultrasonography, a variety of isotopic scans, and positron emission tomography (PET) have supplanted older, more invasive approaches and opened new diagnostic vistas. In light of their capabilities and the rapidity with which they can lead to a diagnosis, it is tempting to order a battery of imaging studies. All physicians have had experiences in which imaging studies revealed findings that led to an unexpected diagnosis. Nonetheless, patients must endure each of these tests, and the added cost of unnecessary testing is substantial. Furthermore, investigation of an unexpected abnormal finding may lead to an iatrogenic complication or to the diagnosis of an irrelevant or incidental problem. A skilled physician must learn to use these powerful diagnostic tools judiciously, always considering whether the results will alter management and benefit the patient.

■ MANAGEMENT OF PATIENT CARE

Team-Based Care Medical practice has long involved teams, particularly physicians working with nurses and, more recently, with physician assistants and nurse practitioners. Advances in medicine

have increased our ability to manage very complex clinical situations (e.g., intensive care units [ICUs], bone marrow transplantation) and have shifted the burden of disease toward chronic illnesses. Because an individual patient may have multiple chronic diseases, they may be cared for by several specialists as well as a primary care physician. In the inpatient setting, care may involve multiple consultants along with the primary admitting physician. Communication through the medical record is necessary but not sufficient, particularly when patients have complex medical problems or when difficult decisions need to be made about the optimal management plan. Physicians should optimally meet face-to-face or by phone to ensure clear communication and thoughtful planning. It is important to note that patients often receive or perceive different messages from various care providers; thus, attempts should be made to provide consistency among these messages to the patient. Management plans and treatment options should be outlined succinctly and clearly for the patient.

Another dimension of team-based care involves allied health professions. It is not unusual for a hospitalized patient to encounter physical therapists, pharmacists, respiratory therapists, radiology technicians, social workers, dietitians, and transport personnel (among others) in addition to physicians and nurses. Each of these individuals contributes to clinical care as well as to the patient's experience with the health care system. In the outpatient setting, disease screening and chronic disease management are often carried out by nurses, physician assistants, or other allied health professionals. The roles and responsibilities of various team members can be confusing for patients. Within reason, different providers should introduce themselves, explain their role, and wear nametags or other identifying information.

The growth of team-based care has important implications for medical culture, student and resident training, and the organization of health care systems. Despite diversity in training, skills, and responsibilities among health care professionals, common values need to be espoused and reinforced. Many medical schools have incorporated interprofessional teamwork into their curricula. Effective communication is inevitably the most challenging aspect of implementing team-based care. While communication can be aided by electronic devices, including medical records, apps, or text messages, it is vitally important to balance efficiency with taking the necessary time to speak directly with colleagues.

The Dichotomy of Inpatient and Outpatient Internal Medicine The hospital environment has undergone sweeping changes over the past few decades. Emergency departments and critical care units have evolved to manage critically ill patients, allowing them to survive formerly fatal conditions. In parallel, there is increasing pressure to reduce the length of stay in the hospital and to manage complex disorders in the outpatient setting. This transition has been driven not only by efforts to reduce costs but also by the availability of new outpatient technologies, such as imaging and percutaneous infusion catheters for long-term antibiotics or nutrition, minimally invasive surgical procedures, and evidence that outcomes often are improved by reducing inpatient hospitalization. Telehealth and remote monitoring tools can also enhance connectivity with patients and reduce visits to the clinic or hospital.

In addition to traditional medical beds, hospitals now encompass multiple distinct levels of care, such as the emergency department, procedure rooms, overnight observation units, critical care units, and palliative care units. A consequence of this differentiation has been the emergence of new specialties (e.g., emergency medicine and end-of-life care) and the provision of in-hospital care by hospitalists and intensivists. Most *hospitalists* are board-certified internists who bear primary responsibility for the care of hospitalized patients and whose work is limited entirely to the hospital setting. The shortened length of hospital stay means that most patients receive only acute care while hospitalized; the increased complexities of inpatient medicine make the presence of an internist with specific training, skills, and experience in the hospital environment extremely beneficial. *Intensivists* are board-certified physicians who are further certified in critical care medicine and who direct and provide care for very ill patients in critical care

units. Clearly, an important challenge in internal medicine today is to ensure the continuity of communication and information flow between a patient's primary care physician and those who are in charge of the patient's hospital care. Maintaining these channels of communication is frequently complicated by patient "handoffs"—i.e., transitions from the outpatient to the inpatient environment, from the critical care unit to a general medicine floor, from a medical to a surgical service and vice versa, from the hospital environment to the recently developed "home hospital" setting (for select patients with adequate home support), and from the hospital or home hospital to the outpatient environment.

The involvement of many care providers in conjunction with these transitions can threaten the traditional one-to-one relationship between patient and primary care physician. Of course, patients can benefit greatly from effective collaboration among a number of health care professionals; however, *it is the duty of the patient's principal or primary physician to coordinate these collaborations and provide cohesive guidance through an illness.* To meet this challenge, primary care physicians must be familiar with the techniques, skills, and objectives of specialist physicians and allied health professionals who care for their patients in the hospital. In addition, primary care physicians must ensure that their patients benefit from scientific advances and the expertise of specialists, both in and out of the hospital. Primary care physicians should explain the role of these specialists to reassure patients that they are in the hands of physicians best trained to manage their current illness. However, the primary care physician should assure patients and their families that decisions are being made in consultation with these specialists. The evolving concept of the "medical home" incorporates team-based primary care with subspecialty care in a cohesive environment that ensures smooth transitions of care.

Mitigating the Stress of Acute Illness Few people are prepared for a new diagnosis of cancer or anticipate the occurrence of a myocardial infarction, stroke, or major accident. The care of a frightened or distraught patient is confounded by these understandable responses to life-threatening events. The physician and other health providers can reduce the shock of life-changing events by providing information in a clear, calm, consistent, and reassuring manner. Often, information and reassurance need to be repeated. Caregivers should also recognize that, for the typical patient, hospital emergency rooms, operating rooms, ICUs, and general medical floors represent an intimidating and often frightening environment. Hospitalized patients find themselves surrounded by air jets, buttons, and glaring lights; invaded by tubes and wires; and beset by the numerous members of the health care team—hospitalists, specialists, nurses, nurses' aides, physician assistants, social workers, technologists, physical therapists, medical students, house officers, attending and consulting physicians, and many others. They may be transported to special laboratories and imaging facilities replete with blinking lights, strange sounds, and unfamiliar personnel; they may be left unattended at times; and they may be obligated to share a room with other patients who have their own health problems. It is little wonder that patients may find this environment bewildering and stressful. The additive effects of an acute illness, unfamiliar environment, multiple medications, and sleep deprivation can lead to confusion or delirium, especially in older hospitalized patients. Physicians who appreciate the hospital experience from the patient's perspective and who make an effort to guide and comfort the patient through this experience may make a stressful situation more tolerable and enhance the patient's chances for an optimal recovery. The advent of home-hospital care represents an attempt to improve the patient experience during uncomplicated acute illness by providing care in the familiar surroundings of the home environment, mitigating the stress of conventional hospitalization and shortening recovery times.

Medical Decision-Making Medical decision-making is a fundamental responsibility of the physician and occurs at each stage of the diagnostic and therapeutic process. The decision-making process involves the ordering of additional tests, requests for consultations, decisions about treatment, and predictions concerning prognosis. This process requires an in-depth understanding of the pathophysiology and natural history of disease. Formulating a differential diagnosis

requires not only a broad knowledge base but also the ability to assess the relative probabilities of various diseases for a given patient. Application of the scientific method, including hypothesis formulation and data collection, is essential to the process of accepting or ruling out a particular diagnosis. Analysis of the differential diagnosis is an iterative process. As new information or test results are acquired, the group of disease processes being considered can be contracted or expanded appropriately. Whenever possible, decisions should be evidence-based, taking advantage of data from rigorously designed clinical trials or objective comparisons of different diagnostic tests. *Evidence-based medicine* stands in sharp contrast to anecdotal experience, which is often biased. Unless attuned to the importance of using larger, objective studies for making decisions, even the most experienced physicians can be influenced to an undue extent by recent encounters with selected patients. Evidence-based medicine has become an increasingly important part of routine medical practice and has led to the publication of many useful practice guidelines. It is important to remember, however, that only a fraction of the many decisions made in clinical practice are based on rigorous clinical trial evidence; other guideline recommendations are, therefore, predicated on expert consensus and weaker evidentiary support.

Thus, the importance of evidence-based medicine notwithstanding, much medical decision-making still relies on good clinical judgment, an attribute that is difficult to quantify or even to assess qualitatively. Physicians must use their knowledge and experience as a basis for weighing known factors, along with the inevitable uncertainties, and then making a sound judgment; this synthesis of information is particularly important when a relevant evidence base is not available. Several quantitative tools may be invaluable in synthesizing the available information, including diagnostic tests, Bayes' theorem (the probability of an event predicated on prior knowledge of conditions possibly related to the event), and multivariate statistical models ([Chap. 4](#)). Diagnostic tests serve to reduce uncertainty about an individual's diagnosis or prognosis and help the physician decide how best to manage that individual's condition. The battery of diagnostic tests complements the history and physical examination. The accuracy of a particular test is ascertained by determining its sensitivity (true-positive rate) and specificity (true-negative rate), as well as the predictive value of a positive and a negative result. [See Chap. 4 for a more thorough discussion of decision-making in clinical medicine.](#)

Practice Guidelines Many professional organizations and government agencies have developed formal clinical-practice guidelines to aid physicians and other caregivers in making diagnostic and therapeutic decisions that are evidence-based, cost-effective, and most appropriate to a particular patient and clinical situation. As the evidence base of medicine increases, guidelines can provide a useful framework for managing patients with particular diagnoses or symptoms. Clinical guidelines can protect patients—particularly those with inadequate health care benefits—from receiving substandard care. These guidelines also can protect conscientious caregivers from inappropriate charges of malpractice and society from the excessive costs associated with the overuse of medical resources. There are, however, caveats associated with clinical-practice guidelines since they tend to oversimplify the complexities of medicine. Furthermore, groups with different perspectives may develop divergent recommendations regarding issues as basic as the need for screening of women by mammography or of men with serum prostate-specific antigen (PSA) measurements. Finally, guidelines, as the term implies, do not—and cannot be expected to—account for the uniqueness of each individual and their illness. The physician's challenge is to integrate into clinical practice the useful recommendations offered by experts without accepting them blindly or being inappropriately constrained by them.

Precision Medicine The concept of *precision* or *personalized medicine* reflects the growing recognition that diseases once lumped together can be further stratified based on genetic, biomarker, phenotypic, and/or psychosocial characteristics that distinguish a given patient from other patients with similar clinical presentations. Inherent in this concept is the goal of targeting therapies in a more specific way to improve

clinical outcomes for the individual patient and minimize unnecessary side effects for those less likely to respond to a particular treatment. In some respects, precision medicine represents the evolution of clinical practice guidelines, which are usually developed for populations of patients or a particular diagnosis (e.g., hypertension, thyroid nodule). As the pathobiology, prognosis, and treatment responses of subgroups within these diagnoses become better understood (i.e., through refined genomic analysis or enhanced deep phenotyping), the relevant clinical guidelines incorporate progressively more refined recommendations for individuals within these subgroups. The role of precision medicine is best illustrated for cancers in which genetic testing is able to predict responses (or the lack thereof) to targeted therapies ([Chap. 78](#)). Increasingly, biomarkers, including assessment of specific genetic alterations, are being used to define more precisely subsets of disease and identify therapeutic approaches tailored to the cellular pathobiology. Effective management often relies on the detection of biomarker expression. The validity of such tests is certified by the U.S. Food and Drug administration based on the collection of data from clinical trials. One can anticipate similar applications of precision medicine in pharmacogenomics, immunologic disorders, and diseases in which biomarkers can predict treatment responses. [See Chap. 5 for a more thorough discussion of precision medicine.](#)

Evaluation of Outcomes Clinicians generally use *objective* and readily measurable parameters to judge the outcome of a therapeutic intervention. These measures may oversimplify the complexity of a clinical condition as patients often present with a major clinical problem in the context of multiple complicating background illnesses. For example, a patient may present with chest pain and cardiac ischemia, but with a background of chronic obstructive pulmonary disease and renal insufficiency. For this reason, outcome measures, such as mortality, length of hospital stay, or readmission rates, are typically risk-adjusted. An important point to remember is that patients usually seek medical attention for *subjective* reasons; they wish to obtain relief from pain, to preserve or regain function, and to enjoy life. The components of a patient's health status or quality of life can include bodily comfort, capacity for physical activity, personal and professional function, sexual function, cognitive function, and overall perception of health. Each of these important domains can be assessed through structured interviews or specially designed questionnaires. Such assessments provide useful parameters by which a physician can judge patients' subjective views of their disabilities and responses to treatment, particularly in chronic illness. The practice of medicine requires consideration and integration of both objective and subjective outcomes.

Many health systems use survey and patient feedback data to assess qualitative features such as patient satisfaction, access to care, and communication with nurses and physicians. In the United States, HCAHPS (Hospital Consumer Assessment of Healthcare Providers and Systems) surveys are used by many systems and are publicly reported. Social media is also being used to assess feedback in real time as well as to share patient experiences with health care systems, potentially enriching the information available for use in medical decisions.

Errors in the Delivery of Health Care A series of reports from the Institute of Medicine (now the National Academy of Medicine [NAM]) called for an ambitious agenda to reduce medical error rates and improve patient safety by designing and implementing fundamental changes in health care systems ([Chap. 7](#)). It is the responsibility of hospitals and health care organizations to develop systems to reduce risk and ensure patient safety. Medication errors can be reduced through the use of ordering systems that rely on electronic processes or, when electronic options are not available, that eliminate misreading of handwriting, or that highlight important drug interactions. Whatever the clinical situation, it is the physician's responsibility to use powerful therapeutic measures wisely, with due regard for their beneficial actions, potential dangers, and cost. Implementation of infection control systems, enforcement of hand-washing protocols, and careful oversight of antibiotic use can minimize the complications of nosocomial infections. Central-line infection rates and catheter-associated urinary tract infections have been dramatically reduced at

many centers by careful adherence of trained personnel to standardized protocols for introducing, maintaining, and removing central lines and urinary catheters, respectively. Rates of surgical infection and wrong-site surgery can likewise be reduced by the use of standardized protocols and checklists. Falls by patients can be minimized by judicious use of sedatives and appropriate assistance with bed-to-chair and bed-to-bathroom transitions. Taken together, these and other measures are saving thousands of lives each year.

Electronic Medical Records Both the growing reliance on computers and the strength of information technology now play central roles in medicine, including efforts to reduce medical errors. Laboratory data are accessed almost universally through computers. Many medical centers now have electronic medical records (EMRs), computerized order entry, and bar-coded tracking of medications. Some of these systems are interactive, sending reminders or warning of potential medical errors.

EMRs offer rapid access to information that is invaluable in enhancing health care quality and patient safety, including relevant data, historical and clinical information, imaging studies, laboratory results, and medication records. These data can be used to monitor and reduce unnecessary variations in care and to provide real-time information about processes of care and clinical outcomes. Ideally, patient records are easily transferred across the health care system; however, technological limitations and concerns about privacy and cost continue to limit broad-based use of EMRs in many clinical settings.

For all of the advantages of EMRs, they can create distance between the physician and patient if care is not taken to preserve face-to-face contact. EMRs also require training and time for data entry. Many providers spend significant time entering information to generate structured data and to meet billing requirements. They may feel pressured to take short cuts, such as “cutting and pasting” parts of earlier notes into the daily record, thereby increasing the risk of errors. EMRs also structure information in a manner that disrupts the traditional narrative flow across time and among providers. These features, which may be frustrating for some providers, must be weighed against the advantages of ready access to past medical history, imaging, laboratory data, and consultant notes. Furthermore, the effort, time, and attention needed to maintain and utilize the EMR have led to a growing sense of dissatisfaction among physicians, lessening professional and personal well-being as a result. Clearly, this is an area of daily practice that requires improvement both for the delivery of safe and optimal care and physician wellness.

It is important to emphasize that information technology is merely a tool and can never replace the clinical decisions that are best made by the physician. Clinical knowledge and an understanding of a patient's needs, supplemented by quantitative tools, still represent the best approach to decision-making in the practice of medicine.

THE PATIENT-PHYSICIAN RELATIONSHIP

The significance of the intimate personal relationship between physician and patient cannot be too strongly emphasized, for in an extraordinarily large number of cases both the diagnosis and treatment are directly dependent on it. One of the essential qualities of the clinician is interest in humanity, for the secret of the care of the patient is in caring for the patient.

—Francis W. Peabody, October 21, 1925,
Lecture at Harvard Medical School

Physicians must never forget that patients are individuals with problems that all too often transcend their physical complaints. They are not “cases” or “admissions” or “diseases.” Patients do not fail treatments; treatments fail to benefit patients. This point is particularly important in this era of high technology in clinical medicine. Most patients are anxious and fearful. Physicians should instill confidence and offer reassurance, but they must never come across as arrogant, patronizing, impatient, or hurried. A professional attitude, coupled with warmth and openness, can do much to alleviate anxiety and to encourage patients to share all aspects of their medical history. Empathy and compassion are the essential features of a caring physician. The

physician needs to consider the setting in which an illness occurs—in terms not only of patients themselves but also of their familial, social, and cultural backgrounds. The ideal patient-physician relationship is based on thorough knowledge of the patient, mutual trust, and the ability to communicate.

Informed Consent The fundamental principles of medical ethics require physicians to act in the patient's best interest and to respect the patient's autonomy. Both principles are reflected in the process of informed consent. Patients are required to sign consent forms for most diagnostic or therapeutic procedures. Many patients possess limited medical knowledge and must rely on their physicians for advice. Communicating in a clear and understandable manner, physicians must fully discuss the alternatives for care and explain the risks, benefits, and likely consequences of each alternative. The physician is responsible for ensuring that the patient thoroughly understands these risks and benefits; encouraging questions is an important part of this process. It may be necessary to go over certain issues with the patient more than once. This is the very definition of *informed consent*. Complete, clear explanation and discussion of the proposed procedures and treatment can greatly mitigate the fear of the unknown that commonly accompanies hospitalization. Often the patient's understanding is enhanced by repeatedly discussing the issues in an unthreatening and supportive way and answering new questions that occur to the patient as they arise. Continuing efforts to educate the patient are essential. Patients are frequently inhibited from understanding by the fear of an uncertain future and potential impact of the illness on themselves and their families. Clear communication can also help alleviate misunderstandings in situations where complications of intervention occur. Special care should also be taken to ensure that a physician seeking a patient's informed consent has no real or apparent conflict of interest.

Approach to Grave Prognoses and Death No circumstance is more distressing than the diagnosis of an incurable disease, particularly when premature death is inevitable. What should the patient and family be told? What measures should be taken to maintain life? What can be done to optimize quality of life?

Transparency of information, delivered in an appropriate manner, is essential in the face of a terminal illness. Even patients who seem unaware of their medical circumstances, or whose family members have protected them from diagnoses or prognoses, often have keen insights into their condition. They may also have misunderstandings that can lead to additional anxiety. The patient must be given an opportunity to speak with the physician and ask questions. A wise and insightful physician uses such open communication as the basis for assessing what the patient wants to know and when they want to know it. On the basis of the patient's responses, the physician can assess the most appropriate time and pace for sharing information. Ultimately, the patient must understand the expected course of the disease so that appropriate plans and preparations can be made. The patient should participate in decision-making with an understanding of the goal of treatment (palliation) and its likely effects. The patient's religious beliefs should be taken into consideration. Some patients may find it easier to share their feelings about death with their physician, nurses, or members of the clergy than with family members or friends.

The physician should provide or arrange for emotional, physical, and spiritual support, and must be compassionate, unhurried, and open. In many instances, there is much to be gained by the laying on of hands. Pain should be controlled adequately, human dignity maintained, and isolation from family and close friends avoided. These aspects of care tend to be overlooked in hospitals, where the intrusion of life-sustaining equipment can detract from attention to the individual person and encourage concentration instead on the life-threatening disease, against which the battle ultimately will be lost in any case. In the face of terminal illness, the goal of medicine must shift from *cure* to *care* in the broadest sense of the term. *Primum succurrere*, first to help, is a guiding principle. In offering care to a dying patient, a physician should be prepared to provide information to family members and deal with their grief and sometimes their feelings of guilt or even anger. It is important for the physician to assure the family that everything

reasonable is being done. A substantial challenge in these discussions is that the physician often does not know exactly how to gauge the prognosis. In addition, various members of the health care team may offer different opinions. Good communication among providers is essential so that consistent information is provided to patients. This is especially important when the best path forward is uncertain. Advice from experts in palliative and terminal care should be sought whenever appropriate to ensure that clinicians are not providing patients with unrealistic expectations. **For a more complete discussion of end-of-life care, see Chap. 13.**

Maintaining Humanism and Professionalism Many trends in the delivery of health care tend to make medical care feel impersonal. These trends, some of which have been mentioned already, include (1) vigorous efforts to reduce the escalating costs of health care; (2) the growing number of managed-care programs, which are intended to reduce costs but where the patient may have little choice in selecting a physician; (3) increasing reliance on technological advances and computerization; (4) the need for numerous physicians and other health professionals to be involved in the care of most patients who are seriously ill; and (5) the growth of telemedicine and virtual encounters that place physical limitations on patient interactions.

In light of these changes in the medical care system, it is a major challenge for physicians to maintain the *humane* aspects of medical care. The American Board of Internal Medicine, working together with the American College of Physicians–American Society of Internal Medicine and the European Federation of Internal Medicine, has published a *Charter on Medical Professionalism* that underscores three main principles in physicians' contract with society: (1) the primacy of patient welfare, (2) patient autonomy, and (3) social justice. While medical schools appropriately place substantial emphasis on professionalism, a physician's personal attributes, including integrity, respect, and compassion, also are extremely important.

Availability to the patient, expression of sincere concern, willingness to take the time to explain all aspects of the illness, and a nonjudgmental attitude when dealing with patients whose cultures, lifestyles, attitudes, and values differ from those of the physician are just a few of the characteristics of a humane physician. Every physician will, at times, be challenged by patients who evoke strongly negative or positive emotional responses. Physicians should be alert to their own reactions to such situations and should consciously monitor and control their behavior so that the patient's best interest remains the principal motivation for their actions at all times. Many organizations use implicit bias training to assist with these skills.

Another important aspect of patient care involves an appreciation of the patient's "quality of life," a subjective assessment of what each patient values most. This assessment requires detailed, sometimes intimate knowledge of the patient, which usually can be obtained only through candid, unhurried, and often repeated conversations. Time pressures will always threaten these interactions, but they should not diminish the importance of understanding and seeking to fulfill the priorities of the patient. The patient encounter should be focused on the patient and not on distractions such as data available in the EMR displayed on the office or bedside computer.

■ EXPANDING FRONTIERS IN MEDICAL PRACTICE

The Era of "Omics" In the spring of 2003, announcement of the complete sequencing of the human genome officially ushered in the genomic era. However, even before that landmark accomplishment, the practice of medicine had been evolving as a result of insights into both the human genome and the genomes of a wide variety of microbes. The clinical implications of these insights are illustrated by the complete genome sequencing of H1N1 influenza virus in 2009 and even faster sequencing of COVID-19 in early 2020, leading to the swift development and dissemination of safe and effective vaccines. Today, gene expression profiles are being used to guide therapy and inform prognosis for a number of diseases, and genotyping is providing a new means to assess the risk of certain diseases as well as variations in response to a number of drugs. Despite these advances, the use of

complex genomics in the diagnosis, prevention, and treatment of disease is still in its early stages. The task of physicians is complicated by the fact that phenotypes generally are determined not by genes alone but by the complex interactions among genes and gene products and by the interplay of genetic and environmental factors.

Rapid progress is also being made in other areas of molecular medicine. *Epigenetics* is the study of alterations in chromatin and histone proteins and methylation of DNA sequences (often caused by environmental factors) that influence gene expression (**Chap. 497**). Every cell of the body has identical DNA sequences; the diverse phenotypes a person's cells manifest are, in part, the result of epigenetic regulation of gene expression. Epigenetic alterations are associated with a number of cancers and other diseases. *Proteomics* is the study of the entire library of proteins made in a cell or organ and the complex relationship of these proteins to disease. This huge repertoire of proteins is generated through alternate splicing of genes, posttranslational processing, and posttranslational modifications that often have unique functional consequences. The presence or absence of particular proteins in the circulation or in cells is being explored for many diagnostic and disease-screening applications. *Microbiomics* is the study of the resident microbes in humans and other mammals, which together compose the microbiome. The human haploid genome has ~23,000 genes, whereas the microbes residing on and in the human body encompass more than 3–4 million genes; these resident microbes are likely to be of great significance with regard to health status of the host. Ongoing research is demonstrating that the microbes inhabiting human mucosal and skin surfaces play a critical role in maturation of the immune system, in metabolic balance, in brain function, and in disease susceptibility. A variety of environmental factors, including the use and overuse of antibiotics, have been tied experimentally to substantial increases in disorders such as obesity, metabolic syndrome, atherosclerosis, and immune-mediated diseases in both adults and children. *Metagenomics*, of which microbiomics is a part, is the genomic study of environmental species that have the potential to influence human biology directly or indirectly. An example is the study of exposures to microorganisms in farm environments that may be responsible for the lower incidence of asthma among children raised on farms. *Metabolomics* is the study of the range of metabolites in cells or organs and the ways they are altered in disease states. The aging process itself may leave telltale metabolic footprints that allow the prediction (and possibly the prevention) of organ dysfunction and disease. It seems likely that disease-associated patterns will be found in lipids, carbohydrates, membranes, mitochondria and mitochondrial function, and other vital components of cells and tissues. *Exposomics* is the study of the exposome—i.e., the environmental exposures such as smoking, sunlight, diet, exercise, education, and violence that together have an enormous impact on health. All of this new information represents a challenge to the traditional reductionist approach to medical thinking. The variability of results in different patients, together with the large number of variables that can be assessed, creates challenges in identifying preclinical disease and defining disease states unequivocally. Accordingly, the tools of *systems biology* and *network medicine* are being applied to the enormous body of information ("big data") now obtainable for every patient and may eventually provide new approaches to classifying, diagnosing, treating, and preventing disease. **For a more complete discussion of a complex systems and network science approach to human disease, see Chap. 499.**

The rapidity of these advances may seem overwhelming to practicing physicians; however, physicians have an important role to play in ensuring that these powerful technologies and sources of new information are applied judiciously to patient care. Since omics are evolving so rapidly, physicians and other health care professionals must engage in continuous learning so that they can apply this new knowledge to the benefit of their patients' health and well-being. Genetic testing requires wise counsel based on an understanding of the value and limitations of the tests as well as the implications of their results for specific individuals. **For a more complete discussion of genetic testing, see Chap. 480.**

The Globalization of Medicine Physicians should be cognizant of diseases and health care services beyond local boundaries. Global

travel has critical implications for disease spread, and it is not uncommon for diseases endemic to certain regions to be seen in other regions after a patient has traveled to and returned from those regions. The outbreak of Zika virus infections in the Americas is a cogent example of this phenomenon. In addition, factors such as wars, the migration of refugees, travel patterns, and the impact of increasing climate extremes are contributing to changing disease profiles worldwide. Patients have broader access to unique expertise or clinical trials at distant medical centers, even those in other countries, and the cost of travel may be offset by the quality of care at those distant locations. As much as any other factor influencing global aspects of medicine, the Internet has transformed the transfer of medical information throughout the world. This change has been accompanied by the transfer of technological skills through telemedicine and international consultation—for example, interpretation of radiologic images and pathologic specimens. **For a complete discussion of global issues, see Chap. 485.**

Medicine on the Internet On the whole, the Internet has had a positive effect on the practice of medicine; through personal computers, a wide range of information is available to physicians and patients almost instantaneously at any time and from anywhere in the world. This medium holds enormous potential for the delivery of current information, practice guidelines, state-of-the-art conferences, journal content, textbooks (including this text), and direct communications with other physicians and specialists, expanding the depth and breadth of information available to the physician regarding the diagnosis and care of patients. Medical journals are now accessible online, providing rapid sources of new information. By bringing them into direct and timely contact with the latest developments in medical care, this medium also serves to lessen the information gap that has hampered physicians and health care providers in remote areas.

Patients, too, are turning to the Internet in increasing numbers to acquire information about their illnesses and therapies and to join Internet-based support groups. Patients often arrive at a clinic visit with sophisticated information about their illnesses. In this regard, physicians are challenged in a positive way to keep abreast of the latest relevant information while serving as an information “editor” as patients navigate this seemingly endless source of information, the accuracy and validity of which are not uniform.

A critically important caveat is that virtually anything can be published on the Internet, with easy circumvention of the peer-review process that is an essential feature of academic publications. Both physicians and patients who search the Internet for medical information must be aware of this danger. We have recently seen the adverse consequences of misinformation and disinformation (often in service of some political agenda) in social media. It is difficult to calculate the harm done by the widespread deceptions that prevented perhaps millions of people from accepting life-saving preventions like the COVID-19 vaccine. Notwithstanding this limitation, appropriate use of the Internet is revolutionizing information access for physicians and patients, and in this regard, the Internet represents a remarkable resource that was not available to practitioners a generation ago.

Public Expectations and Accountability The general public’s level of knowledge and sophistication regarding health issues has grown rapidly over the past few decades. As a result, expectations of the health care system in general and of physicians in particular have risen. Physicians are expected to master rapidly advancing fields (the *science* of medicine) while considering their patients’ unique needs (the *art* of medicine). Thus, physicians are held accountable not only for the technical aspects of the care they provide but also for their patients’ satisfaction with the delivery and costs of care.

In many parts of the world, physicians increasingly are expected to account for the way in which they practice medicine by meeting certain standards prescribed by federal and local governments. The hospitalization of patients whose health care costs are reimbursed by the government and other third parties is subjected to utilization review. Thus, a physician must defend the cause for and duration of a patient’s hospitalization if it falls outside certain “average” standards. Authorization for reimbursement increasingly is based on documentation of the

nature and complexity of an illness, as reflected by recorded elements of the history and physical examination. A growing “pay-for-performance” movement seeks to link reimbursement to quality of care. The goal of this movement is to improve standards of health care and contain spiraling health care costs. In many parts of the United States, managed (capitated) care contracts with insurers have replaced traditional fee-for-service care, placing the onus of managing the cost of all care directly on the providers and increasing the emphasis on preventive strategies. In addition, physicians are expected to give evidence of their current competence through mandatory continuing education, patient record audits, maintenance of certification, and relicensing.

Medical Ethics and New Technologies The rapid pace of technological advances has profound implications for medical applications that go far beyond the traditional goals of disease prevention, treatment, and cure. Cloning, gene therapy, gene editing, human–computer interfaces, nanotechnology, and use of targeted therapies have the potential to modify inherited predispositions to disease, select desired characteristics in embryos, augment “normal” human performance, replace failing tissues, and substantially prolong life span. Given their unique training, physicians have a responsibility to help shape the debate on the appropriate uses of and limits placed on these new technologies and to consider carefully the ethical issues associated with the implementation of such interventions. As medicine becomes more complex, shared decision-making is increasingly important, not only in areas such as genetic counseling and end-of-life care but also in diagnostic and treatment options.

Learning Medicine More than a century has passed since the publication of the Flexner Report, a seminal study that transformed medical education and emphasized the scientific foundations of medicine as well as the acquisition of clinical skills. In an era of burgeoning information and access to medical simulation and informatics, many schools are implementing new curricula that emphasize lifelong learning and the acquisition of competencies in teamwork, communication skills, system-based practice, and professionalism. The tools of medicine also change continuously, necessitating formal training in the use of EMRs, large datasets, ultrasound, robotics, and new imaging techniques. These and other features of the medical school curriculum provide the foundation for many of the themes highlighted in this chapter and are expected to allow physicians to progress, with experience and learning over time, from competency to proficiency to mastery.

At a time when the amount of information that must be mastered to practice medicine continues to expand, increasing pressures both within and outside of medicine have led to the implementation of restrictions on the amount of time a physician-in-training can spend in the hospital and in clinics. Because the benefits associated with continuity of medical care and observation of a patient’s progress over time were thought to be outstripped by the stresses imposed on trainees by long hours and by fatigue-related errors, strict training parameters were introduced by the Accreditation Council for Graduate Medical Education (ACGME) to reduce fatigue and enhance learning. The impact of these changes is still being assessed, and there are continual efforts to optimize training and move toward competency-based endpoints. An unavoidable by-product of fewer hours at the bedside is an increase in the number of “handoffs” of patient responsibility from one physician to another. It is imperative that these transitions of responsibility be handled with care and thoroughness, with all relevant information exchanged and acknowledged. These issues highlight the challenge our profession has in establishing a reliable measure of physician effectiveness.

The Physician as Perpetual Student From the time physicians graduate from medical school, it becomes all too apparent that this milestone is symbolic and that they must embrace the role of a “perpetual student.” This realization is at the same time exhilarating and anxiety-provoking. It is exhilarating because physicians can apply constantly expanding knowledge to the treatment of their patients; it is anxiety-provoking because physicians realize that they will never know as much as they want or need to know. Ideally, physicians will